Orders over the complete selected 2-volcano

This separately frozen phase measures the cost of obtaining curve and point orders. It changes no original solver input, timing or checkpoint. All 46 curves and 9200 P/Q checks are included.

Exact curve cardinality	72057979853368804=4×18014494963342201, independently recomputed on every curve
Native / PARI point orders	9200 / 9200, all equal to 18014494963342201; every result matched
Reference point-order arithmetic per call	12381 field multiplications including squares; 147 inversions. Full square/add/group/branch counters accompany the data.
Curve-count CPU / elapsed range	164–218 ms CPU; 171–237 ms elapsed. One call per curve.
Timing scope	Intel Xeon Platinum 8573C, four-CPU cgroup, 16 GiB; L0 with concurrent BSGS. Native is serial; GP default internal thread count is 5. No isolated or fastest-tool comparison.

What is and is not charged

Native point-order timing starts with supplied exact #E and its factorization. It divides candidate orders using scalar-multiplication tests. Membership and independent validation are outside the interval. The 200 point checks per curve intentionally repeat each generator ten times because that is the fixed workload; they are not 200 independently sampled generators. Square counts are a subset of multiplication-including-square and must not be added twice.

PARI separately records cardinality, factorization, group structure and 200 point-order calls per curve. Factorization and group-structure stages follow ellcard and can reuse cached curve information. GP has 1 ms timer resolution: a returned zero is below resolution, not zero actual cost. Whole-process wall, CPU, peak RSS, page faults and context switches are retained in validation/order_*_resources.json.

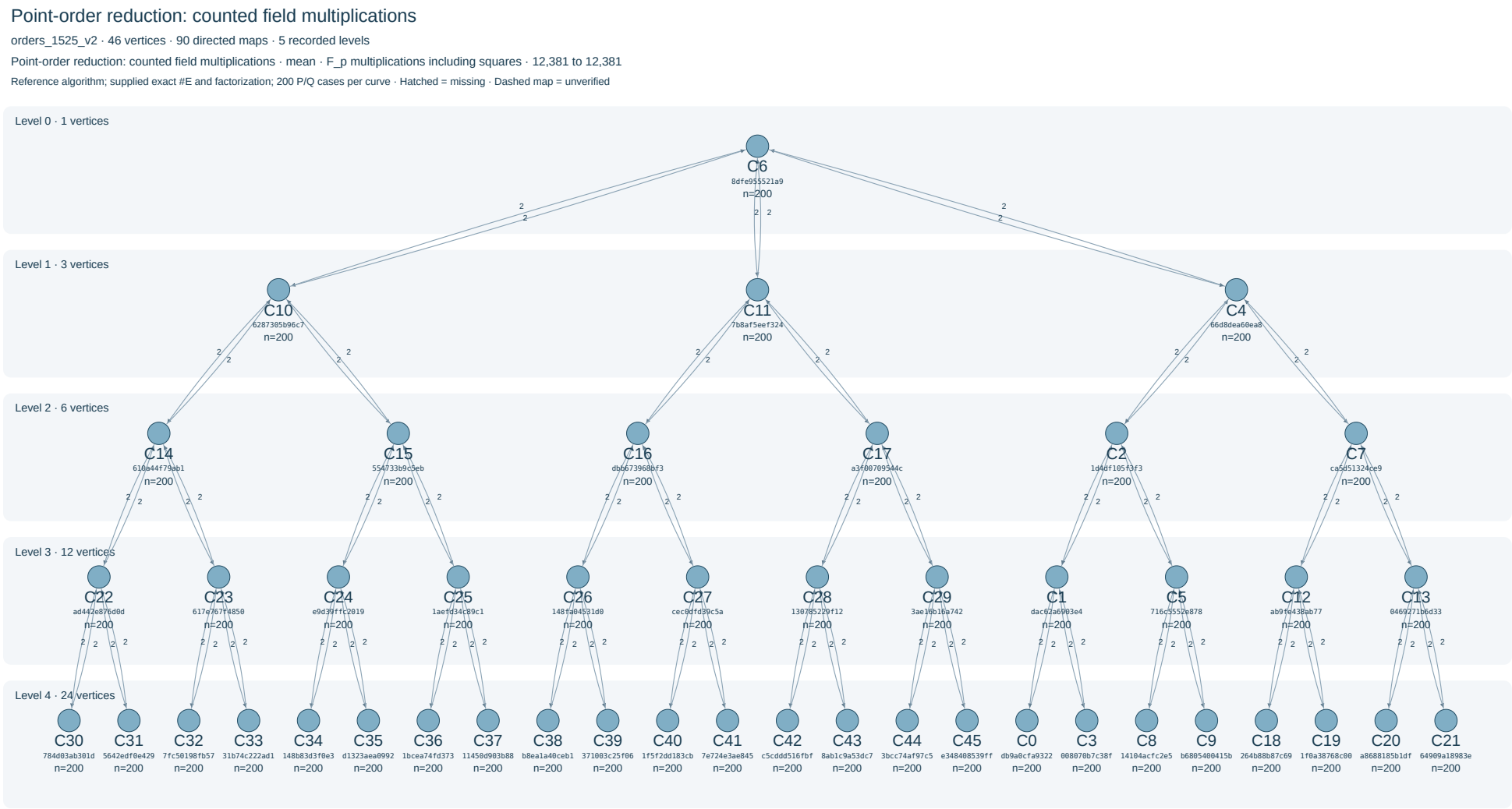
Interpretation and remaining scope

Full-group invariant factors differ between cyclic order 4r and Z/2 × Z/(2r), but every benchmark P and Q still has prime order r. Every transported degree-2 map is injective on that subgroup. Computing or checking the order is part of experiment setup, not a replacement for solving the ECDLP. Identical arithmetic counts with varying elapsed time do not establish different hardness.

These point-order counters do not fill the missing rho/BSGS internal field counters. Full random-corpus IC recovery remains unresolved, and BSGS completion is tracked separately. Earlier snapshots and failures are preserved. Sources: ORDER_PROTOCOL.md, ORDER_FROZEN.json, source/order_measure.cpp, source/order_setup.mjs and this report generator. Exact curves and complete maps: data/family.json and data/formal_maps.jsonl.

Point-order reduction: counted field multiplications

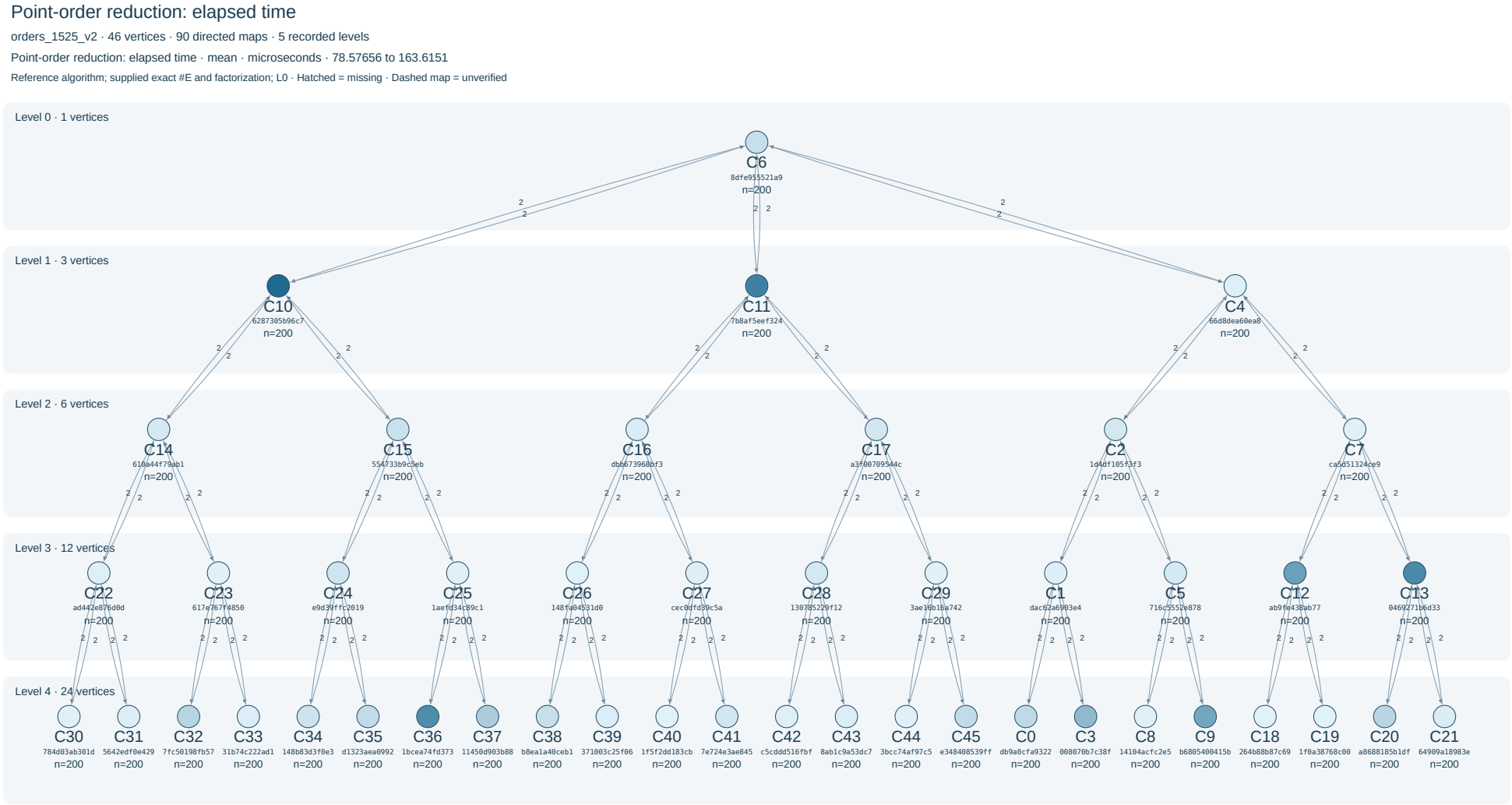
Reference algorithm; supplied exact #E and factorization; 200 P/Q cases per curve. Unit: F_p multiplications including squares. Exact statistics and curve identities: point_order_operations/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Point-order reduction: elapsed time

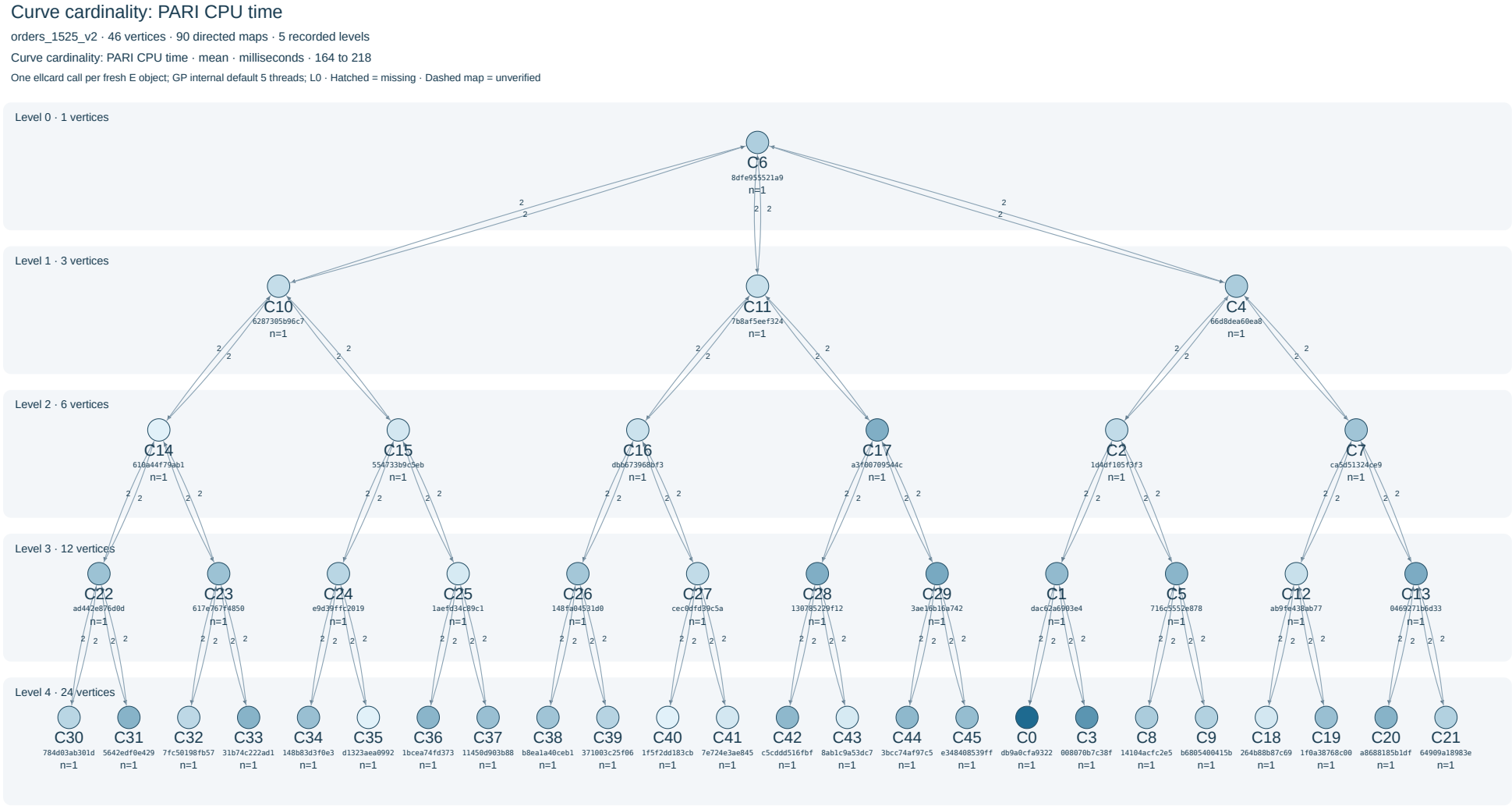
Reference algorithm; supplied exact #E and factorization; L0. Unit: microseconds. Exact statistics and curve identities: point_order_wall/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Curve cardinality: PARI CPU time

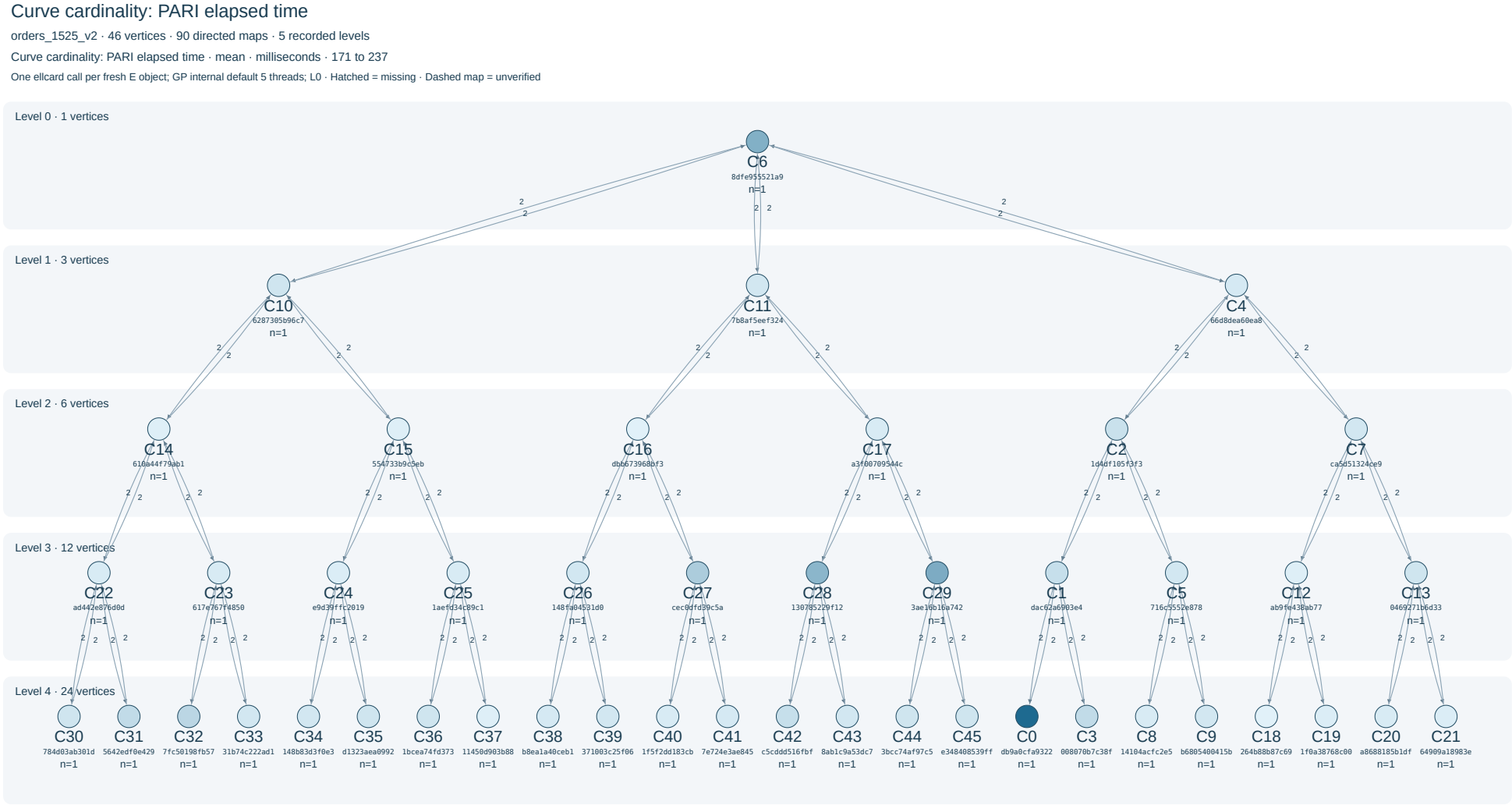
One ellcard call per fresh E object; GP internal default 5 threads; L0. Unit: milliseconds. Exact statistics and curve identities: curve_cardinality_cpu/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Curve cardinality: PARI elapsed time

One ellcard call per fresh E object; GP internal default 5 threads; L0. Unit: milliseconds. Exact statistics and curve identities: curve_cardinality_wall/graph_data.json.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.